

Revised for the R1 revision of HELIYON-D-26-00946. Entries shown in blue were added or rewritten in this revision; entries in black are unchanged from the original submission. Two points on how to read it. (1) The four items previously marked only “N/A” (Items 14, 15, 21 and 22) now carry an explicit justification, as requested by Reviewer 2. (2) Locations are given as section, table and figure references rather than page numbers: the revision adds substantial text and pagination will change on recompilation, so section references are the stable and exact pointer. Page numbers can be inserted at proof stage if the journal requires them.

PRISMA 2020 Main Checklist

Topic	No.	Item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Page 1, Title
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist	Page 1, Abstract. See also the PRISMA 2020 for Abstracts checklist (second table in this document).
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Pages 1–3, Section 1
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page 4, Section 2
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Section 2.1, paragraphs “Operational definition of clinical isomorphism” and “Study-design criteria” (both added in R1). The isomorphism criterion satisfied by each included study is recorded in the deposited extraction dataset.
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Section 2.1, paragraph “Search reporting” (added in R1): Scopus only; English only; 2000 to January 2023; date last searched January 2023; reference lists of all included articles inspected manually, contributing 4 further articles. An empirical PubMed recall check is reported in Section 4.4.
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Table 2 (all five search strings verbatim, with Scopus field tags, Boolean operators, the language limit and per-string record counts) and Section 2.1, paragraph “Search reporting”. Also reproduced on the Search_Strategy worksheet of the deposited extraction dataset.

Topic	No.	Item	Location where item is reported
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Section 2.1 and Figure 1. Screening was not performed in independent duplicate; this is stated in Section 4.4, paragraph "Single-extractor data collection".
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Section 2.1. Extraction was performed primarily by the first author and reviewed by the co-authors; not performed in independent duplicate (Section 4.4).
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Section 2.1. The complete list of extracted data items is the column set of the Included_Studies worksheet of the deposited extraction dataset.
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Section 2.1. No assumptions were made about missing or unclear information; fields not reported by a study are left empty in the deposited dataset rather than imputed.

Topic	No.	Item	Location where item is reported
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Section 2.1, paragraph “Quality and risk-of-bias assessment” (rewritten in R1). No formally validated instrument was used; the three pre-specified appraisal dimensions and their scoring rules are stated there. Rationale and consequences: Section 4.4, paragraph “No formally validated risk-of-bias instrument”.
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Not applicable — no effect measures were pooled. The model performance metrics extracted for each study are defined in Section 2.1 and their distribution is reported in Table 4.
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item 5)).	Section 2.1, paragraph “Synthesis methods” (added in R1).
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Section 2.1, paragraph “Synthesis methods” (added in R1).
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Section 2.1, paragraph “Synthesis methods”; tabulation and display conventions for Tables 3 and 4 are given in Section 3.
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Section 2.1, paragraph “Synthesis methods”, which states the rationale for structured narrative synthesis rather than meta-analysis. No meta-analysis was performed; see also Section 4.4, paragraph “No meta-analysis”.

Topic	No.	Item	Location where item is reported
Reporting bias assessment	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Not applicable — no quantitative synthesis was performed, so no meta-regression or other statistical exploration of heterogeneity was possible. Sources of heterogeneity are discussed narratively in Section 3 and Section 4.4.
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Not applicable — no quantitative synthesis was performed, so no sensitivity analysis of a pooled estimate was possible.
	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Not applicable — no quantitative synthesis was performed. Assessment of risk of bias due to missing results (e.g. funnel-plot asymmetry, small-study-effect tests) requires pooled comparable effect estimates, which do not exist for this review. The substantive risk is addressed narratively in Section 4.4, paragraph “No certainty-of-evidence grading and no reporting-bias assessment”.
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Not applicable — no outcome was pooled, and GRADE domains (imprecision, inconsistency, indirectness relative to a pooled estimate) are undefined in the absence of a quantitative synthesis. Certainty considerations are addressed narratively in Section 4.4.
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Section 2.1 (narrative rewritten in R1 to reproduce every stage of Figure 1) and Figure 1.
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Figure 1 and Section 2.1, with the number excluded at each stage and the reason. A record-level list of the 103 full-text exclusions with individual reasons was not retained; this is stated in Section 4.4.
Study characteristics	17	Cite each included study and present its characteristics.	Section 3, Figures 2–3 and Table 4. Full per-study characteristics for all 111 studies are given in the Included_Studies worksheet of the deposited extraction dataset.
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Section 3, paragraph “Critical appraisal of the included studies”, and Table 3 (both added in R1). Per-study ratings for all 111 studies are in the Quality_Assessment worksheet of the deposited extraction dataset.

Topic	No.	Item	Location where item is reported
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Section 3 (narrative, by study) and Table 4 (distribution of best reported values by metric and category). Per-study values are in the deposited extraction dataset.
Results of syntheses	20a	For each synthesis, briefly summarize the characteristics and risk of bias among contributing studies.	Section 3, paragraph “Critical appraisal of the included studies”, and Table 3.
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Not applicable — no statistical synthesis was performed. The narrative synthesis and its summary tabulations are reported in Section 3, Tables 3 and 4.
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Not applicable — no statistical synthesis was performed, so no statistical investigation of causes of heterogeneity was possible.
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Not applicable — no statistical synthesis was performed, so no sensitivity analysis was possible.
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Not applicable — see Item 14. No quantitative synthesis, therefore no formal reporting-bias assessment. The direction of the risk is stated in Section 4.4: AI performance results are subject to strong publication and reporting bias, and the near-ceiling metrics in Table 4 are the expected signature of it.
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Not applicable — see Item 15. Certainty of the body of evidence is discussed narratively in Section 4.4 rather than graded.
DISCUSSION			

Topic	No.	Item	Location where item is reported
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Section 4 (Discussion), Sections 4.1–4.3 (the three-phase roadmap) and Section 5 (Conclusions).
	23b	Discuss any limitations of the evidence included in the review.	Section 4.4 (Limitations), in particular the paragraphs “Dependence on proxy-domain evidence and scarcity of melioidosis-specific data” and “No certainty-of-evidence grading and no reporting-bias assessment”.
	23c	Discuss any limitations of the review processes used.	Section 4.4 (Limitations), in particular “Single-database search” (which now reports an empirical PubMed recall check), “No formally validated risk-of-bias instrument”, “Single-extractor data collection”, “No meta-analysis” and “Search currency”.
	23d	Discuss implications of the results for practice, policy, and future research.	Sections 4.1–4.3 and 4.5, and Section 5 (Conclusions).
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	The review was NOT registered in PROSPERO or any other register. Stated in Section 2.1.
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	No protocol was prepared for public access. Planning followed the PRISMA-P 2015 extension, but the protocol was neither registered nor published; stated in Section 2.1.
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Not applicable — no registered or published protocol exists; therefore, no amendments to record.
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Abstract (funding sentence) and the Acknowledgment section.
Competing interests	26	Declare any competing interests of review authors.	Conflicts of Interest section.

Topic	No.	Item	Location where item is reported
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Data Availability section (rewritten in R1). The complete extraction dataset for all 111 included studies — harmonized extraction fields, per-study quality ratings underlying Table 3, performance values underlying Table 4, verbatim search strings and per-stage screening counts — is deposited in an open repository under a CC-BY license https://github.com/suppawong/Melioidosis-AI-SLR , together with this checklist. No analysis code was generated by this review.

PRIMSA Abstract Checklist

Topic	No.	Item	Reported?
TITLE			
Title	1	Identify the report as a systematic review.	Yes
BACKGROUND			
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	Yes
METHODS			
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	Yes
Information sources	4	Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.	Yes — Scopus; searched to January 2023.
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	Partly — the abstract states that a quality assessment was performed to evaluate the technical rigor and validation methods of the included models, but does not specify the appraisal dimensions. These are given in Section 2.1 and their results in Table 3.
Synthesis of results	6	Specify the methods used to present and synthesize results.	Yes
RESULTS			
Included studies	7	Give the total number of included studies and participants and summarize relevant characteristics of studies.	Yes — 111 included studies. Total number of participants is not applicable: the unit of analysis in this review is the study/model, not the patient.
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured).	Yes
DISCUSSION			

Topic	No.	Item	Reported?
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (e.g. study risk of bias, inconsistency and imprecision).	Yes — the abstract reports the limitations of the evidence base explicitly: “The evidence base is, however, methodologically weak: only 18 studies (16.2%) drew on two or more independent data providers, and 27% reported accuracy without a discrimination metric, so published performance should be read as an upper bound.” Expanded in Section 3 (Table 3) and Section 4.4.
Interpretation	10	Provide a general interpretation of the results and important implications.	Yes
OTHER			
Funding	11	Specify the primary source of funding for the review.	The funding is acknowledged separately.
Registration	12	Provide the register name and registration number.	No register name or registration number applies.

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